Cybersecurity Project 2 – Basic Cryptography (Hashing)

Objective

To understand and use basic hashing techniques using the Linux terminal, and learn how hashes can be used for file integrity and password storage.

Tools Needed

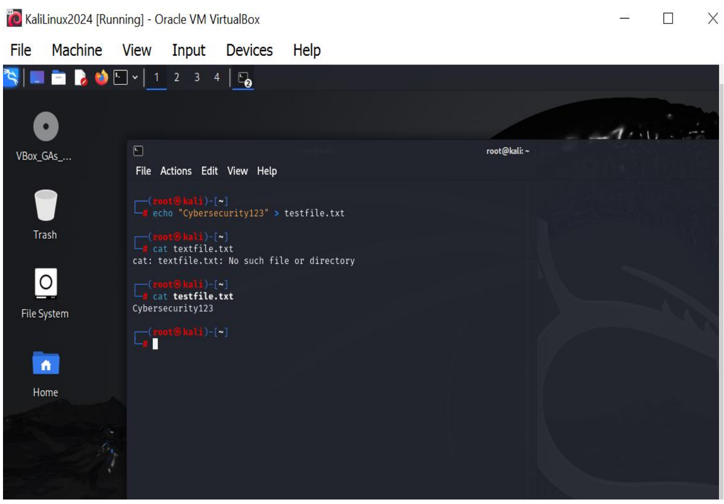
- ✓ Kali Linux VM (from your home lab setup)
- ✓ Terminal (pre-installed)

Step 1: Create a Text File

Open the Terminal in Kali.

echo "Cybersecurity123" > testfile.txt
This creates a file called testfile.txt and writes the word Cybersecurity123 into it.

Your terminal after running the command and confirming file creation with:cat **testfile.txt**

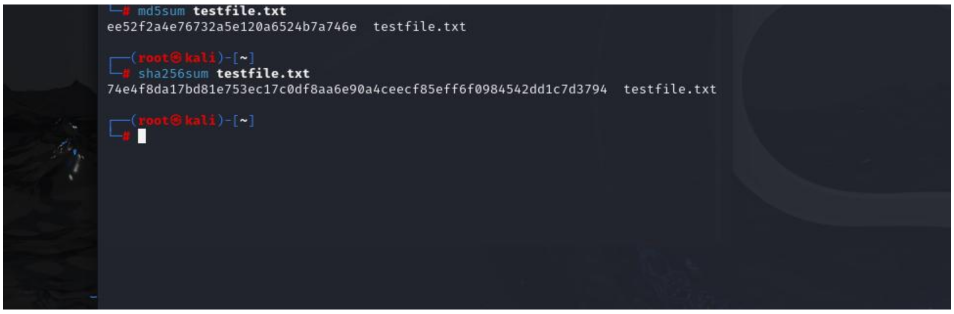


Step 2: Generate MD5 Hash

md5sum testfile.txt
This gives you an MD5 hash a kind of fingerprint of the file. If the file changes, the hash will change too.

Step 3: Generate SHA256 Hash

sha256sum testfile.txt
This gives you a **stronger** hash (SHA256). Still a fingerprint, but harder to fake or break.



Step 4: Change the File and Check Again

echo "I changed it!" >> testfile.txt

Now recheck the hashes:

md5sum testfile.txt
sha256sum testfile.txt

You added new text to the file. So the content changed → the hash will also change.

Step 5: (Optional) Try with an Empty File

touch empty.txt
md5sum empty.txt
sha256sum empty.txt
Even an empty file has a unique hash! This shows how everything has a hash.